CSC_52081_EP Advanced Machine Learning and Autonomous Agents
Quizz week 8 (Representation learning and fairness)

Deadline: Wednesday March 19, 2025, 12h00 Paris time (midday, not midnight)

Instructions: Submit your answers by putting them at the end of the lab08's notebook, following the instructions that you see there. The deadline for the quiz is the same as the deadline for the code submission, there is nothing separate to do for the quiz than put the answers at the end of the notebook and submit the notebook by the deadline.

For each question, you need to check the correct statements. There may be 0, 1, or multiple correct statements; checked incorrect statements bring negative score.

Question 1

Consider, in a variational autoencoder, the standard expression

$$\mathcal{L}(\phi, \theta) = \mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)] - KL(q_\phi(z|x)||p(z)),$$

where $p(z)$ is a standard normal distribution, $p_\theta(x|z)$ is the conditional density of the data conditioned on the latent variables z indexed by the model parameter θ, and $q_\phi(z|x)$ is a normal distribution distribution on z given x indexed by parameter ϕ.

a. The term $\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)]$ is the data log-likelihood function.

b. The term $\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)]$ can be computed through a closed-form expression.

c. The term $\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)]$ is usually estimated through a Monte-Carlo approach by drawing a single sample of z.

d. The term $KL(q_\phi(z|x)||p(z))$ can be computed through a closed-form expression.

e. The term $KL(q_\phi(z|x)||p(z))$ is usually estimated through a Monte-Carlo approach by drawing a single sample of z.

Question 2

Consider a dataset of vectors in $\mathbb{R}^d$ with dimension $d \in \mathbb{N}$. We compare two unsupervised learning algorithms with the same value of $k < d$:

- PCA, with k principal components;
- An autoencoder with a single hidden layer of k and only linear functions.

a. Both algorithms find the same linear subspace of dimension k on which the data is projected.

b. The distance between a data point x and its projection $\hat{x}$ (or reconstruction in the autoencoder language) is the same for both algorithms.

c. Both algorithms end up with exactly the same "code" (i.e., value of the k-dimensional representation z) for every data point x.

d. The autoencoder is trained using an eigenvalue decomposition of the data covariance matrix.

Question 3

Consider a classification scenario with only a sensitive feature $A \in \{a, b\}$, and a label $Y \in \{0, 1\}$, defined by the joint distribution:

$$
\begin{aligned}
P(Y = 0, A = a) &= 0.40 \\
P(Y = 1, A = a) &= 0.10 \\
P(Y = 0, A = b) &= 0.30 \\
P(Y = 1, A = b) &= 0.20
\end{aligned}
$$

Consider a classifier such that $\hat{Y} = 0$ with probability 0.7 independently of A and Y. The score R is identical to $\hat{Y}$.

a. The classifier satisfies the fairness notion of sufficiency.

b. The classifier satisfies the fairness notion of independence.

c. The classifier satisfies the fairness notion of separation.

d. In this particular scenario, it is possible to construct a classifier that satisfies simultaneously the fairness notions of independence and sufficiency.

e. In this particular scenario, it is impossible to construct a classifier that satisfies the fairness notion of sufficiency.